# BIPLOB ROY

Data Engineer · Data Warehouse Specialist · AI Enthusiast

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## PROFESSIONAL SUMMARY

Results-driven Data Engineer with strong expertise in data warehouse design, ETL pipeline development, and large-scale data migration. Proven experience building **Snowflake OLAP systems** on MS SQL Server and transforming complex transactional data into analytics-ready structures. Started in .NET backend, now specializing in Data Engineering and AI-driven solutions (**ML.NET, ONNX**). Delivered international healthcare and transport systems (Zambia) with end-to-end data architecture ownership.

## CORE SKILLS

- **Data Engineering & Warehousing:** ETL Pipelines (SQL, Python, SSIS), Snowflake Schema Design, Fact/Dimension Modeling, OLTP → OLAP Transformation, Large-scale Data Migration, Apache Superset
- **Databases:** Microsoft SQL Server (Advanced — OLAP, Agent Jobs, Performance Tuning), PostgreSQL (Advanced — Functions, Reporting), MongoDB, MySQL
- **Backend & Distributed:** C#, ASP.NET MVC, Web API, .NET 8, Microservices Architecture, Kafka, Background Services
- **AI / Data Science & Tools:** ML.NET (Intent Detection, Chatbots), ONNX Model Integration, Apache Spark (ML & Big Data), Docker, Git, SQL Server Agent, Airflow, SSIS

## CAREER HIGHLIGHTS

- Architected **SCRS** — a centralized OLAP data warehouse serving the entire healthcare reporting stack; replaced legacy OLTP reporting with optimized Snowflake schema, significantly cutting report query execution time.
- Engineered a **real-time TSRS Data Transport Pipeline** syncing transactional data every second into the OLAP warehouse — enabling live SLA dashboards via Apache Superset.
- Owned end-to-end **multi-system healthcare migration** (SmartCare Plus → Care Pro → SCRS) and designed the complete **PostgreSQL data architecture** for BTMS Zambia covering OLTP & analytics.
- Built an **AI chatbot (R&D)** with ML.NET + ONNX intent detection, including a dynamic retraining pipeline and intent-based function calling for the full ticket-purchase lifecycle.

## PROFESSIONAL EXPERIENCE

### Data Engineer
Nov 2025 — Present

*Excel Technologies Ltd., Dhaka*

- Designed **SCRS** (Smart Care Reporting Service) using Snowflake OLAP architecture on MS SQL Server; built fact & dimension tables optimized for high-performance reporting queries.
- Led end-to-end data migration pipeline **SmartCare Plus → Care Pro → SCRS**; developed custom ETL pipelines using raw SQL scheduled via SQL Server Agent (self-designed jobs).
- Built **TSRS Data Transport Pipeline** syncing transactional data every second from User & Incident DBs into the TSRS OLAP Data Warehouse.
- Designed PostgreSQL architecture for **BTMS (Zambia)**; developed Python ETL pipelines and a conversational **ML.NET chatbot** with ONNX intent detection (R&D).

### Junior Software Engineer
Mar 2023 — Oct 2025

*Excel Technologies Ltd., Dhaka*

- Developed backend systems using **ASP.NET MVC & Web API**; designed and optimized SQL Server relational databases.
- Built data-driven modules, integrations, and biometric authentication systems (SecuGen fingerprint verification).

### Junior Software Engineer
Mar 2021 — Mar 2023

*Digital Creation Plan, Khulna*

- Developed **WPF-based applications** and database systems; worked with IoT-integrated real-time data systems; improved database performance through schema optimization.

### System Support Engineer
May 2019 — Jan 2020

*Getco Online Ltd., Dhaka*

- Provided **IT infrastructure** and system support; ensured system reliability and troubleshooting across enterprise environments.

## KEY PROJECTS

**SCRS — Smart Care Reporting Service (OLAP Data Warehouse)** Nov 2025 — Present

*MS SQL Server · Snowflake Schema · SQL Server Agent · OLAP*

— Centralized OLAP data warehouse for healthcare reporting. Snowflake schema with optimized fact/dimension tables; custom ETL via raw SQL + SQL Agent scheduling; migrated Care Pro → SCRS ensuring data consistency.

**Tera Support — Mail Service (Event-Driven)** Nov 2025 — Present

*.NET 8 · Kafka · SMTP/IMAP · Docker*

— Kafka-driven mail service: auto-ticket creation from incoming emails, threaded replies into ticket messages, dynamic receiver lists for incident status updates.

**BTMS AI Chatbot (Conversational AI & Intent Detection)** Nov 2025 — Present

*ML.NET · ONNX · .NET · SQL*

— Conversational AI chatbot with ONNX intent detection for English language. Dynamic training pipeline retrains the model based on updated intent datasets. 10-minute session inactivity manager; intent-based function calling for full ticket-purchase lifecycle.

**Care Pro Data Migration Pipeline** Jan 2024 — Dec 2024

*MS SQL Server · SSIS · SQL*

— Designed migration flow SmartCare Plus → Care Pro → SCRS. Developed ETL logic for data cleaning, transformation, and validation; handled large-scale structured data migration.

**RewardsPipeline (ETL System)** Jun 2023 — Dec 2023

*Python · MongoDB · MS SQL · Docker*

— Automated ETL pipeline using pymongo (MongoDB extraction) and pyodbc (SQL loading). Dockerized the solution for scalability with environment configs for flexibility.

**Earlier Projects — Fingerprint Verification · RobRest IoT Restaurant · DCP Resource Platform** Mar 2021 — Oct 2023

*C# · .NET · SecuGen · WPF · MSSQL · IoT · ASP.NET*

— Biometric authentication system (SecuGen SDK); WPF-based IoT smart restaurant with kitchen automation; internal training & resource platform with authentication, admin control, and file management.

**Tera Support — TSRS Reporting Service** Nov 2025 — Present

*.NET 8 · PostgreSQL · OLAP · Apache Superset*

— High-performance Data Transport Pipeline syncing data every second into TSRS OLAP Warehouse. .NET API for Superset dashboards with advanced SLA filtering and Excel exports.

**Tera Support — Multi-Microservice Platform** Nov 2025 — Present

*Kafka · .NET 8 · Microservices*

— Architected the overarching event-driven architecture connecting User, Incident, CRM, and Device Manager modules. Orchestrated real-time data flow via Kafka.

**Arc One & Ageing Index Pipelines (ETL)** Jan 2025 — Oct 2025

*Python · MySQL · MS SQL · Apache Superset*

— Python-based ETL pipelines migrating data from MySQL to MS SQL Server. Built complex database views to generate analytical reports consumed by Apache Superset dashboards.

**BTMS — Bus Ticket Management System (Zambia)** Mar 2023 — Dec 2024

*PostgreSQL · ASP.NET Web API · SQL Functions*

— Designed complete PostgreSQL database schema for OLTP and reporting/analytics. Built complex SQL functions for reporting and developed APIs for report generation and data access.

**SmartCare Pro Reporting & Database Replication** Mar 2023 — Dec 2023

*SQL Server · SSIS · Replication*

— Migrated transactional data to OLAP structure; implemented transactional replication with secondary reporting database to reduce load on primary and improve availability.

## EDUCATION

**B.Sc. in Computer Science & Engineering** CGPA: 3.26

*Stamford University Bangladesh, Dhaka*

## CERTIFICATIONS

**IBM / Coursera — Data Engineering Professional Certificate (Apr 2026)**

~~C~~apstone Project (Oct 2025) · Machine Learning with Apache Spark (Jul 2025) · ETL & Data Pipelines — Airflow, Kafka (Jan 2025) · Data Warehouse Fundamentals (Feb 2025) · NoSQL Databases — MongoDB (Apr 2025) · Relational Database Administration — DBA (Dec 2024) · Python for Data Engineering (Nov 2024)

**Specialized & Other**

~~G~~enerative AI: Elevate your Data Engineering Career — IBM (Apr 2026) · BI Dashboards with IBM Cognos & Google Looker — IBM (Feb 2025) · Introduction to Big Data with Spark & Hadoop — IBM (Jun 2025) · Data Science Specialization — Coursera (16 Courses) · SQL (Intermediate & Basic) — HackerRank (Mar & Feb 2025) · C (Intermediate & Basic) — HackerRank (Sep 2020) · Leveraging ICT for Growth — George Washington University (Jun 2017)

## CAREER OBJECTIVE

> *To become a* **Data Engineer with AI specialization**, *building scalable data warehouse systems and intelligent analytics solutions for* **banking, financial services, and enterprise platforms**.